



CC AIRWELL

Climate Changer
Qatar



Qatar

CC Airwell

*A new technology – a new awareness –
A new international patent – a circulatory system*

- Water harvesting in the desert – 1,7-billion litres/day – 620-million m³ water/year.
- Desert greening – within 10 years up to 1,000 km² (100,000 ha) around CC Airwell.
- Active climate cooling – "recycle" 600-million m³ hot air/day – *we consume heat.*
- We clean polluted air – as a side effect: 600-million m³ per day

Qatar is possibly the most water-stressed country on Earth. According to a recent report from the U.S. World Resources Institute, it is one of 17 countries classified as suffering from extremely high water stress. Qatar doesn't have enough freshwater resources, and water storage is a major concern. With one CC Airwell we will provide 1,7-billion litres of fresh water per day (additional the groundwater under the forest) and we green the desert (better to say: nature will do it, we give the impulse). Our cold air valley and the first forest ring around which we plant (not single trees with irrigation system) is a powerful impulse. The fast growing forest will spread fast (without extra irrigation – rainforest principle). Once we start, it's not to stop.

Construction costs: 800 Million Euro (incl. land, energy supply and my costs/fees)

Construction time: 18 month

The **Doha** area has an annual average air temperature of 29°C! The annual average relative humidity is 53%. At a temp. of 29°C the dew point is 19°C. One cubic meter of this air contains 17 millilitres of water per m³. The maximum temperature in this region is 41°C and the minimum temperature is 17°C. The dew point for 17°C is 11°C – thus we can work 12 months/year (on the 20 raining days we collect rainwater). We provide springwater quality (highest quality, fresh and cold), no extra water treatment necessary.



CC AIRWELL

Qatar

It's all about the dew point:

At the CC Airwell (an international patent) this waste-product "cold air" is valuable and being very well used; with the right amount/volume (6,000 m³ cold air/second) we specifically blow out 2°C "cold air" (up to minus 4°C) into the hot/warm environment. With this temperature of 2°C – 10°C, the air in the cold-air-valley is far under the dew point and thus creating "ground condensation": fog and drizzle on the 2 km² cold-air-valley – we let it rain.

This "ground condensation" seeps into the soil, re-cools the heated-up soil – and collected in 30 m depth. These 10 drops (per second and m²) of water-harvest adds to a total of up to 1.7 billion litres of water/day – used as drinking water, irrigation water, for forest and wild animals. Additionally, we "consume" the sucked in heat and the compression heat from the cooling system (normally a waste-product – forcing global warming).



DEW POINT at 20°C: 2°C
... at 30°C: 11°C
... at 40°C: 19°C
... at 50°C: 28°C

Qatar

Evaporation:

I am often asked: "What if the drops from the ground-condensation (fog and drizzle) are quickly sucked up by hot air (called "evaporation")". The answer: "It's absolutely impossible". Once we can see fog and drizzle, then we see the proof of a meeting between hot air and cold air. And when there is a dew point, we can see condensation. And this fog and drizzle is also the protection-carpet for all what is below. Not even one drop of water can then be sucked up by the hot air above. Once we see condensation, we can be sure: all water is going down. Once we see nothing, we know there is an "evaporation" (if the air above is warmer) – water moves up into the hot air. Fog and drizzle are only existing because there is a cold air source on the ground (rivulets, rivers, lakes or a valley with shade) and fog and drizzle on the ground disappears, once the cold air source from the ground is ending.

At CC Airwell we provide this cold air source on a constant base (forced with the power of sun and wind). And we are not just blowing out cold air, we are creating a 2 km² cold air valley and a cold and moist soil. Most important, we are "holding" this cold soil and cold air, in our cold air valley. All this can be observed every day here in green Austria (we have many cold air sources) and the same happens above a rainforest. I know, many people in desert countries can't imagine this (no cold air sources), but it's existing and it's just nature.

The challenge

at CC Airwell is to cause "ground condensation" – for this we will find the correct settings. Once we can create ground condensation (fog and drizzle), there can be no more evaporation – the fog layer is a barrier – no sun comes through and no heat package can soak up water from the ground – with fog the many drops on the ground are saved – and will seep into the soil. For a ground condensation, the blown-out air needs 1) the correct air speed at the exit, 2) the correct air temperature and 3) the right air volume and it needs 4) the optimal relative humidity. For the correct humidity we use "spray mist" shortly before the air escapes into the hot environment (for this action we need max. 1 million litres of water per day) – only used when necessary.



CC AIRWELL

Qatar

The size of CC Airwell:

The CC Airwell system cannot be built smaller, otherwise we don't have enough cold air (yes, many consider this a mega waste – but it is not) and thus we would be missing the ground condensation (in this case we could not re-cool the heated soil and therefore we could not transfer heat into the soil) – overall we would harvest much less water (maximum 30 million litres per day). The CC Airwell system also cannot be built larger, because then we would have too much cold air, which would cause conflict with our raw material supplier (hot air).



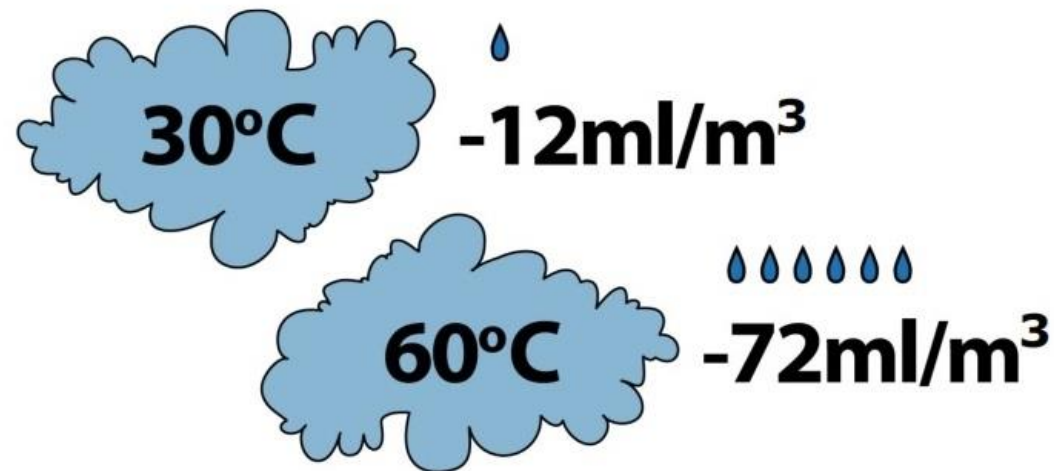
A cold front requires a certain amount of cold air – it is not just about the temperature difference, but also about the volume/amount of the cold air. If we don't dispose enough cold air (with 2°C – 10°C), then nothing happens on the ground. We need at least 6,000 m³ cold air per second (in our 2 km² valley) – this is the only way to achieve ground condensation in our cold air valley – thus fog and drizzle, many drops.



Qatar

It's just 10 drops of water per second and m^2 – Mother nature would say: "Not so much water, not much to see – no big show." But it's much water, because "we let it rain" (ground condensation, fog and drizzle) always on the same place and it's much water because we collect it in 30 m depth.

Once a hot/warm air package and a cooler air package are meeting, we could think "it's just mixing". Yes, it's true, but when the difference of the air-temperatures is too big, then there is a "dew point" and then nothing is mixing, but fog and drizzle is created and to see. In this CC Airwell project, it's all about the dew point – the blown out cold air is far under the dew point and thus creating "ground condensation".



- PS: CC Airwell is no water harvesting machine! CC Airwell is a local climate changing system, it's a local environment cooling system – and one of the results and side effects is ground condensation: Fog and drizzle in our cold-air-valley – thus many drops – and all these drops are seeping into the soil (no evaporation because we have our fog-protection-carpet), re-cooling the soil and collected in 30 m depth.



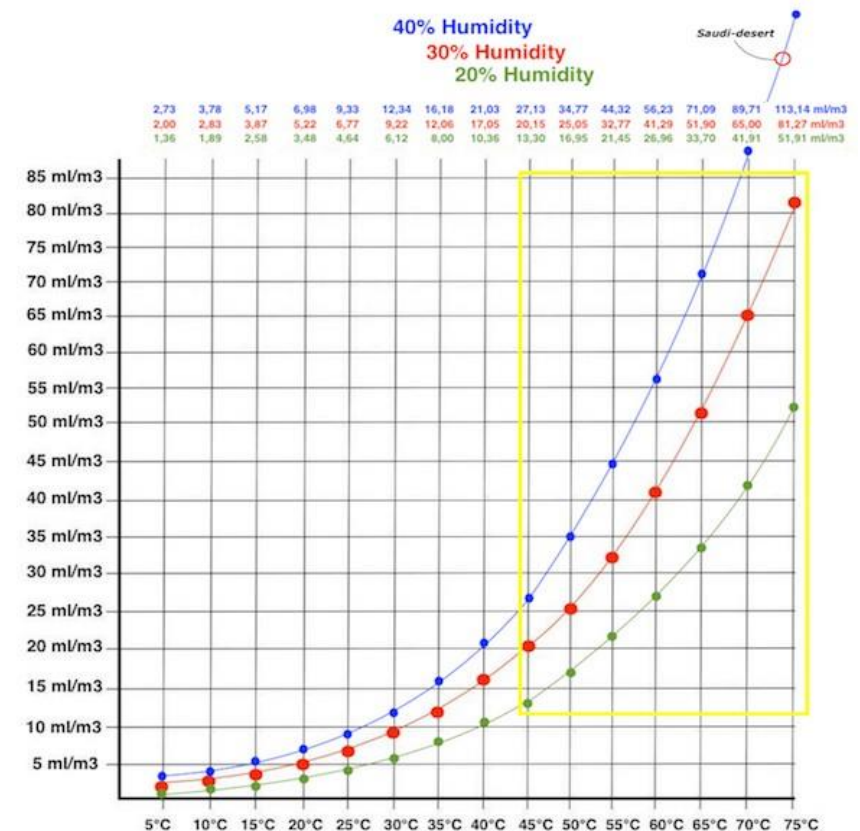
Qatar

Water shortage:

Yes, global air temperatures are rising and Qatar is also facing water shortage, but Qatar area never was losing water, not even one drop – physically absolute impossible. The reason for “water shortage” is not that humans/agriculture use too much water or wasting it – water is under the law of a natural cycle and all the used water goes back into the cycle (ground, rivulets and air). Now in today's times with rising air temperatures worldwide the water still goes back into the natural cycle, but now the most water goes into the hotter air. Once there are no trees and no more big waters (like lakes), the air temperature is jumping up – and a warmer air is capturing much more water (to see in this graph below) – once it's a desert, never again Greenland. CC Airwell will definitely green the desert.

CC Airwell seeks to deliver water in an ethical, sustainable and in a responsible manner. It recognizes the imperative to repair the planet by harnessing heat to produce an amount of water that will be shared by all consumers, i.e., humans, flora and fauna, and in the process create much needed rainforests, it would be incomplete, to only emphasize human consumption needs, to the total exclusion of all else.

The technology consumes heat, an act of counterbalancing the carbon emissions. This technology delivers rain water quality (cleaned while seeping into a depth of 30 m – there we collect it) and the cost of treatment is remarkably reduced, releasing such resources to areas of critical need, such as health. CC Airwell will provide fresh, natural, clean and cool drinking water – without any chemicals, no extra water treatment needed. This project has tremendous potential to put Qatar at the front and centre of the climate change agenda globally.



CC AIRWELL

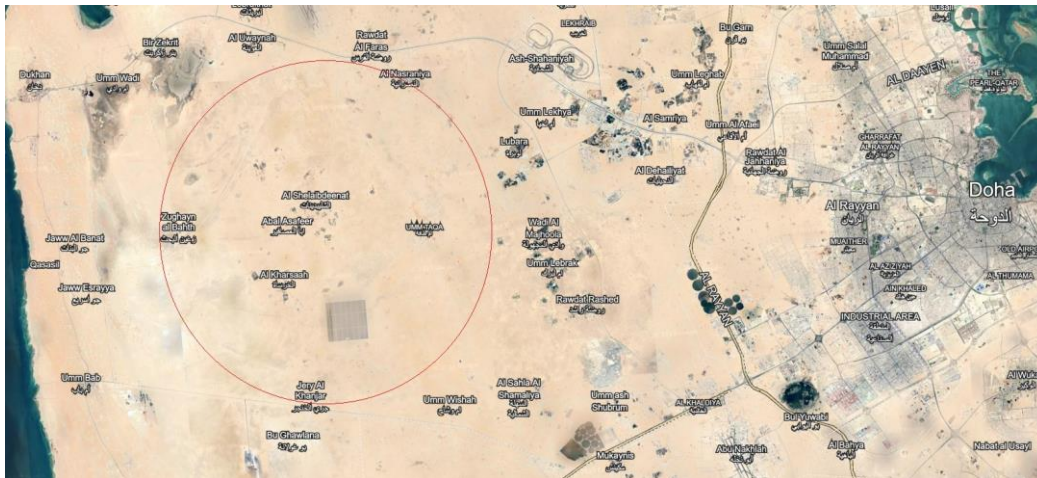
Qatar

Space requirements:

for the CC Airwell plant in Qatar, near Doha we need 100,000 ha desert land (it's a radius of 15 km).

Construction will be in the middle on 10 km²– 1,000 ha.

- 2 km²: CC Airwell system itself (800 m radius) – the cold air valley.
- 2 km²: solar park (3,5 m above the ground).
- 4 km²: forest ring one (within 10 years up to 1,000 km²).
- 2 km²: lakes for wild animals (protecting the plant).



CC AIRWELL

Qatar

The process:

for the CC Airwell plant near Doha (just a suggestion) we need 100,000 ha desert land (a radius of 15 km).

Construction will be in the middle on 1,000 ha.

All starts with caterpillars – *we make a big hole in the desert* – up to 35 m depth, on 200 ha (round, 800 m radius) – then the drainage system (a big water basin/a tube) is laid at a depth of 30 m – then sand comes on it – then the pipes are laid layer by layer – 300 m in length – all must be placed perfect. And finally, we build the cooling tower, the water tank and the pump chamber – after this the interior with electronics and cooling aggregates and the pit ventilators – and on it the telescope towers (for hot air suck in) are built.

Then the energy supply and the test run and the fine adjustment. After this the “village” (for employees and guests), the lake for animals, the water pipe to the next city – the bushes we plant and the first forest ring one.



CC AIRWELL

Qatar

The water harvest:

- Direct condensation water in underground pipe-system 30,000,000 litres of water per day
- Ground condensation – fog and drizzle on 2,000,000 m²
10 drops per second and m² (0,01 litres – 10 ml) are 1,700,000,000 litres of water per day

The supply – the consumption:

- Drinking water, agriculture water – will cost nothing, for free : 1,500,000,000 litres of water per day
- Bushes (in the cold-air-valley), Forest ring one and lake for wild animals 230,000,000 litres of water per day



Qatar

The big picture summary:

Once the first CC Airwell is working, governments in desert countries with water shortage will order CC Airwell because of "drinking water". But for survival on this planet, we humans need the multiple "side effects" of CC Airwell: Consuming heat and CO₂, active climate cooling — no extra costs for climate change — no extra costs for desert greening (side effects), no global warming fight-program is needed and no forced lifestyle change. Within the first 10 years one CC Airwell consumes/captures in total 12-million tons of CO₂ emissions (1,000 km² rainforest within 10 years) — thereafter an average of 2.1 million tons of CO₂/year.

CC Airwell is worldwide the only project, which is consuming Heat and CO₂ — we are using global warming. In reality we can say: "Wow great, the earth temperature is rising every year, so let us finally call this global warming a `valuable commodity` and let us use this heat and transform it into water" (with the CC Airwell we change the aggregate states and make captive water accessible) — and as a side effect we are cooling down our world climate and greening the desert (a big big forest, the best CO₂-consumer).

Rainforests are still being destroyed — that is very sad and seemingly unstoppable. With CC Airwell we don't have any Intention to stop them. We are planting a new rainforest and it's also counteracting the insect and bird extinction — yes, we are creating an insect/animal paradise/island/oasis — we give the vital impulse for this..



CC AIRWELL

Qatar

The Inventor:

Wolfgang Rainer Fuchs

CEO Inventor and international Patent holder (WO 2021/203155 A1) of CC Airwell.

Born 1960 in Austria. This project started in 1981 in India.

Company registration: UID: ATU64078107 – 5020 Salzburg, Austria

E-Mail: fuchs@ccairwell.com – Tel.: +43660/4766478.

Website:

<https://ccairwell.com/>

Videos

<https://www.youtube.com/watch?v=tIR3bF9M-mU>

<https://www.youtube.com/watch?v=jJKeQIIM3io>

<https://www.youtube.com/watch?v=bn-IYXlzYdo&feature=youtu.be>



Maybe "global warming" is the result of our concept "we have to *fight*". And now again we want to *fight* (we humans love to fight) – now fighting against "global warming". In reality we don't have to fight "global warming" – in reality we can use it. Heat is our raw material.



CC AIRWELL